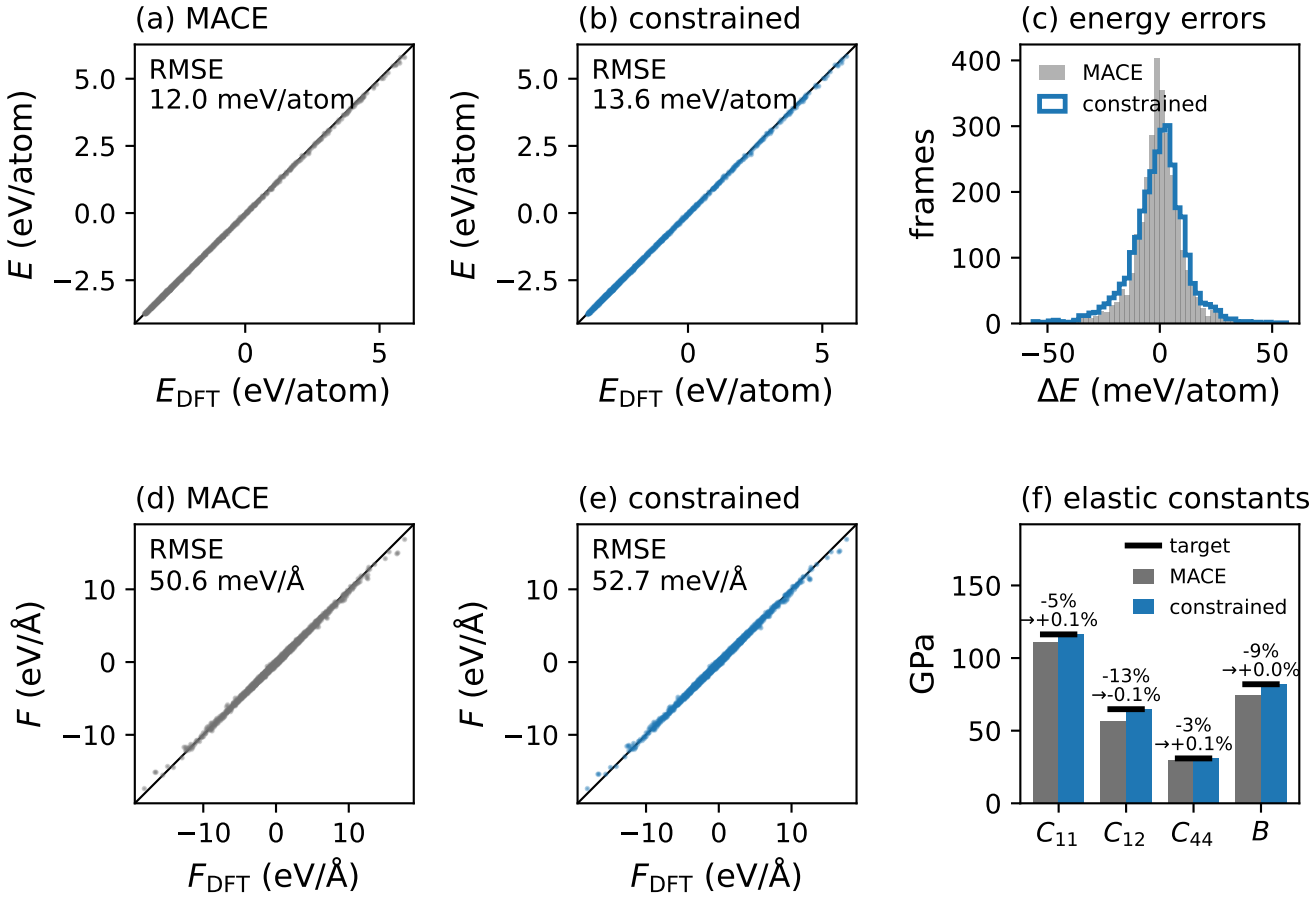
(a) MACE
RMSE
12.0 meV/atom
E (eV/atom)
E_DFT (eV/atom)
(b) constrained
RMSE
13.6 meV/atom
E (eV/atom)
E_DFT (eV/atom)
(c) energy errors
MACE
constrained
frames
ΔE (meV/atom)
(d) MACE
RMSE
50.6 meV/Å
F (eV/Å)
F_DFT (eV/Å)
(e) constrained
RMSE
52.7 meV/Å
F (eV/Å)
F_DFT (eV/Å)
(f) elastic constants
target
MACE
constrained
GPa
-5%
→+0.1%
-13%
→-0.1%
-3%
→+0.1%
-9%
→+0.0%
C11
C12
C44
B